

Python Course for Physicists (And everyone else!)

Chunk 6: Functions: Saving Code

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The Problem: Stupid Copy Paste of Code

- Tasks 5 / Advanced: Check if matrix side lengths are equal

```
matrix_a = [["a", "b", "c"],  
            [1, 9, True]]  
  
matrix_b = [[20, 10.1, 2],  
            [3, 9, 0.1],  
            [10000, 1, 0]]
```

- “Stupid” solution:
→ Copy paste

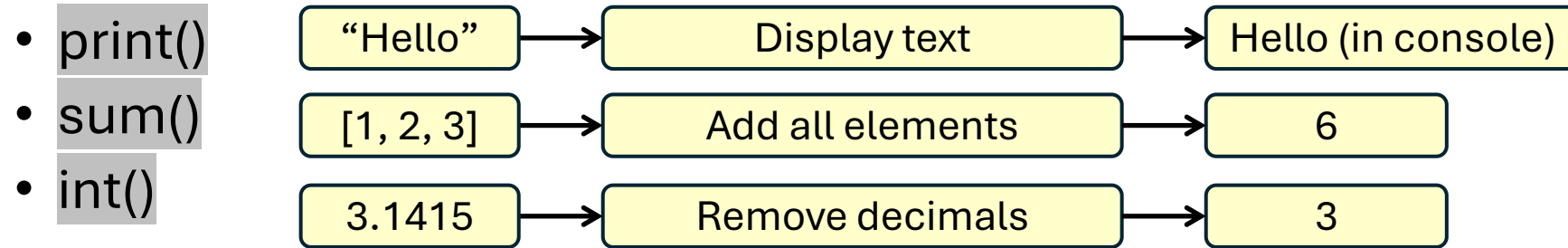
```
if len(matrix_a) == len(matrix_a[0]):  
    print("This is a equally dimensioned matrix")  
else:  
    print("The sides of the matrix have different lengths")
```

+

```
if len(matrix_b) == len(matrix_b[0]):  
    print("This is a equally dimensioned matrix")  
else:  
    print("The sides of the matrix have different lengths")
```

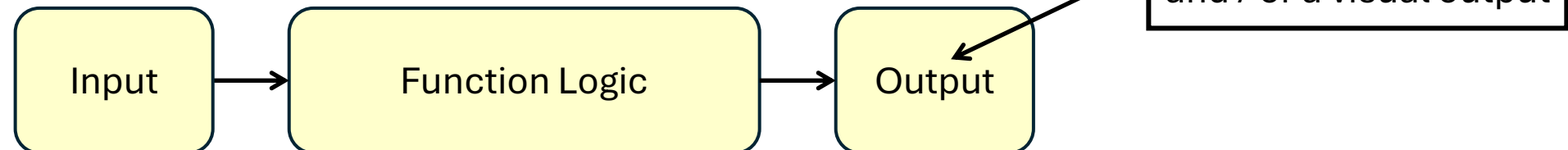
Smart Solution: Functions → Reuse of Code

- Been using functions all the time:



- Logic:

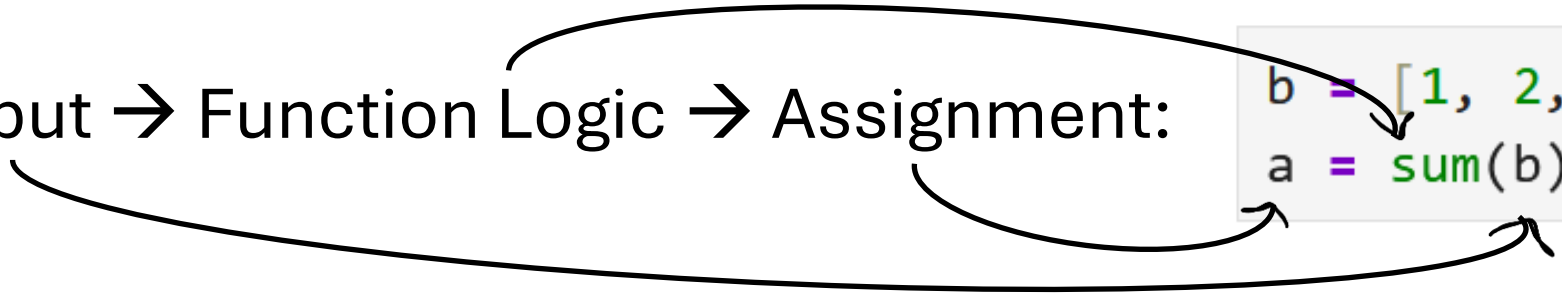
- Input in “()”
- Output: Either text display or saved to a variable



How we use functions

- Input → Function Logic → Assignment:

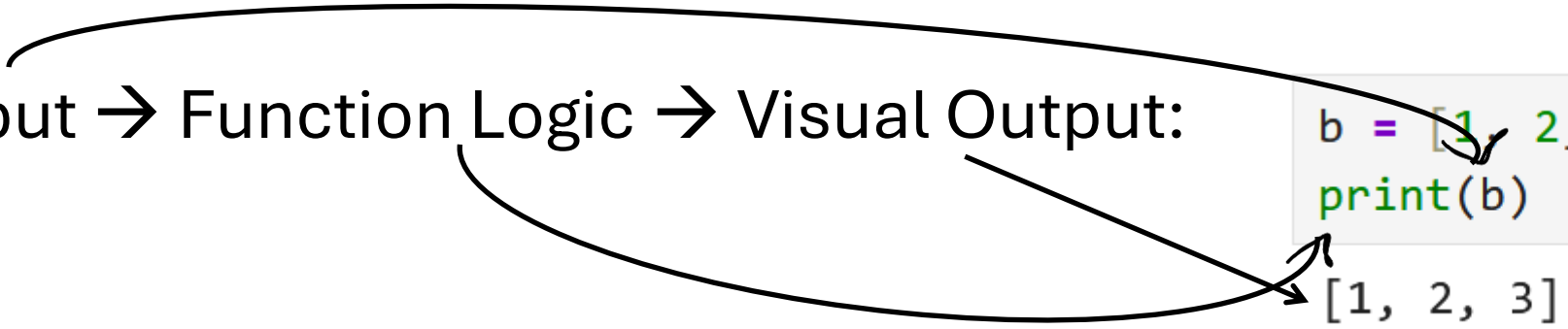
```
b = [1, 2, 3]
a = sum(b)
```



- Input → Function Logic → Visual Output:

```
b = [1, 2, 3]
print(b)
```

[1, 2, 3]



Live Coding: How to Create Functions

- Without Assignment:
(Single Input)

```
def print_double(data):  
    print(data)  
    print(data)  
  
print_double("Hi there")  
  
Hi there  
Hi there
```

- Without Assignment:
(Multiple Input)

```
def calculate_radius(x, y):  
    r = (x**2 + y**2)**(1/2)  
    print(r)  
  
calculate_radius(10, 5)  
  
11.180339887498949
```

Live Coding: How to Create Functions

- With assignment:
(Single return)

```
def calculate_radius(x, y):  
    r = (x**2 + y**2)**(1/2)  
    return r  
  
radius = calculate_radius(10, 5)  
print(radius)  
  
11.180339887498949
```

- With assignment:
(multiple return)

```
from math import sqrt, atan2  
  
def cartesian_to_polar(x, y):  
    r = sqrt(x**2 + y**2)  
    phi = atan2(x, y)  
    return r, phi  
  
r, phi = cartesian_to_polar(10, 5)  
print(r)  
print(phi)  
  
11.180339887498949  
1.1071487177940904
```



[<https://pt.memedroid.com/memes/detail/3846326/Spoiler-alert>]

Comma
separated return

- Function exits after return (even if there is code)

Off topic: f-strings

- Inserting a lot of variables in a string with “+” is tedious:

```
age = 101
name = "Einstein"
fav_food = "Icecream"

print("I am " + name + " and " + str(age) + " years of age.")

I am Einstein and 101 years of age.

print(f"I am {name} and {age} years of age")
```

- Alternative with format():

```
print("I am {} and {} years of age".format(name, age))

I am Einstein and 101 years of age
```

- Alternative with f-string:

```
print(f"I am {name} and {age} years of age")

I am Einstein and 101 years of age
```

Winner 🏆

Good Luck with your Tasks 😊

Programmers looking
at programming memes

